TSPI Round 3 — Complete Domain Expert Response for the Development Team
Based on the Latest Official “39 Axes, 12 Domains, and 3 Keys” Master
This document supersedes all previous answers that conflict with the latest official TSPI framework master provided by the system owner.
The development team should now treat the latest file titled:
“last 39 Axes, 12 Domains, 3 Keys”
as the sole canonical framework for:
Axis names and numbering
Domain assignments
Key hierarchy
Axis-to-network reasoning
Module-to-axis mapping
NSS and scoring logic
Clinical reports
Learning architecture
The Round 3 questions correctly identified that conflicting definitions of Axes 35–39 prevented reliable downstream development. The decision below resolves that blocker.
TSPI_Open_Questions_Round3_For_Experts_260714_075658 (1).pdf
Official TSPI Framework Hierarchy
The confirmed structural hierarchy is:
3 Keys
→ 9 Restoration Steps
→ 12 Domains
→ 39 Biological Axes
→ 180 Living Biological Networks
→ Approximately 200 Phytochemical Modules / Products
The levels have different purposes:
Keys describe the highest-order biological capacities.
Steps describe the restoration sequence.
Domains group major biological territories.
Axes are stable assessment and scoring dimensions.
Networks are the living mechanistic reasoning layer.
Modules are the actual TSPI products used to support network harmonization.
A module is a product. A biological network is not a product and is not interchangeable with a module.
PART 1 — Blocking Questions
Q1. Which 39-Axis dictionary is official?
Developer question
Previous documents gave incompatible definitions for Axes 35–39.
One version stated:
A37 = Protein Quality Control System
A38 = Protein Clearance and Proteotoxic Stress
A39 = Cognitive-Emotional Loop Axis, or CELA
The later Master Dictionary stated:
A35 = Visceral Organ Functional Reserve

A36 = Local Organ and Tissue Interface
A37 = Oncologic Cell Fate
A38 = Tumor Microenvironment and Metastatic Dynamics
A39 = Genomic Regulation Homeostasis
The developers correctly noted that both definitions cannot be used at the same time. The distinction affects Axis scoring, NSS, module matching, dosing logic and every report produced by the engine.
Official answer
Use the latest official 39-Axis Master Dictionary supplied in the file:
“last 39 Axes, 12 Domains, 3 Keys.”
It is the only canonical Axis dictionary.
All earlier material that defined A37–A39 as Protein Quality Control, Protein Clearance and CELA must be marked:
DEPRECATED
SUPERSEDED_BY_LATEST_AXIS_MASTER
The official definitions are listed below.
Official 12 Domains and 39 Biological Axes
Domain 1 — Immune–Infection Governance
Axis
Official English name
A1
Systemic Inflammatory Load
A2
Immune Homeostasis
A3
Pathogen Defense
A4
Post-Infectious Persistence
Domain 2 — Metabolic–Energy Core
Axis
Official English name
A5
Glucose–Insulin Regulation
A6
Lipid–Lipotoxicity
A7
Nutrient Sensing
A8
Mitochondrial Network
A9
Oxidative Stress and Redox Balance
A10
Proteostasis–Autophagy–Lysosome
Domain 3 — Genomic Programming
Axis
Official English name
A11
DNA Repair and Genomic Stability

A12
Epigenetic Regulation
A13
Longevity Signaling
A14
Senescence and SASP
Domain 4 — Detoxification and Cellular Defense
Axis
Official English name
A15
Detoxification Capacity
Domain 5 — Digestive–Gut Ecosystem
Axis
Official English name
A16
Digestive–Absorptive Function
A17
Microbiome Ecology
A18
Gut Barrier and Mucosal Integrity
A19
Colon Biofilm and Mucosal Burden
Domain 6 — Vascular–Circulation
Axis
Official English name
A20
Endothelial Function
A21
Vascular Remodeling and Microcirculation
A22
Thrombosis–Hemostasis
A23
Hemodynamics and Arterial Elasticity
Domain 7 — Repair–Regeneration
Axis
Official English name
A24
Wound Healing
A25
Fibrosis and ECM Remodeling
A26
Stem Cell and Hematopoiesis
A27
Glycation and Carbonyl Stress
Domain 8 — Musculoskeletal and Structural Integrity
Axis
Official English name
A28

Muscle Protein Turnover
A29
Bone Remodeling
A30
Connective Tissue Integrity
Domain 9 — Endocrine Network
Axis
Official English name
A31
Female Hormone Network
A32
Male Androgen Network
A33
Thyroid–Adrenal–Growth Axis
Domain 10 — Neurological Regulation
Axis
Official English name
A34
Neuro–Sleep–Stress–Brain Clearance
Domain 11 — Organ Resilience
Axis
Official English name
A35
Visceral Organ Functional Reserve
A36
Local Organ and Tissue Interface
Domain 12 — Oncology and System-Level Regulation
Axis
Official English name
A37
Oncologic Cell Fate
A38
Tumor Microenvironment and Metastatic Dynamics
A39
Genomic Regulation Homeostasis
Where does Proteostasis belong?
Proteostasis remains under:
A10 — Proteostasis–Autophagy–Lysosome
A10 includes, at minimum:
Protein folding
Chaperone activity
Endoplasmic reticulum stress
Unfolded protein response
Autophagic flux
Lysosomal clearance
Proteasomal degradation
Protein aggregation
Proteotoxic burden

Damaged-organelle clearance
Therefore, Protein Quality Control and Protein Clearance must not be recreated as A37 and A38. That would duplicate the scope already represented by A10.
Sub-axes may later represent the internal components of A10, for example:
A10A — Protein Folding and Chaperone Capacity
A10B — ER Stress and Unfolded Protein Response
A10C — Autophagy–Lysosome Clearance
A10D — Protein Aggregation and Proteotoxic Burden
These are sub-axis details, not additional Parent Axes.
Where does CELA belong?
CELA remains an important part of TSPI, but it is not Axis 39 in the official Master.
CELA should be represented as a named Living Biological Network within the 180-Network layer.
Recommended canonical description:
CELA = Cognitive–Emotional Loop Architecture Network
CELA represents persistent or recurrent cognitive-emotional patterns and their bidirectional relationships with:
Autonomic regulation
Neuroendocrine signaling
Sleep and circadian regulation
Immune activity
Systemic inflammation
Microbiome ecology
Mitochondrial function
Vascular function
Genomic and epigenetic regulation
Its primary Parent Axis should be:
A34 — Neuro–Sleep–Stress–Brain Clearance
It may also maintain secondary relationships with other Axes, including:
A1 — Systemic Inflammatory Load
A2 — Immune Homeostasis
A8 — Mitochondrial Network
A12 — Epigenetic Regulation
A17 — Microbiome Ecology
A20 — Endothelial Function
A33 — Thyroid–Adrenal–Growth Axis
A39 — Genomic Regulation Homeostasis
This distinction is important:
CELA is a network-level construct.
A34 and A39 are stable Parent Axes.
The AI may reason through CELA, but must never rename CELA as A39.
Required implementation action
Freeze the latest Axis Master and attach a framework version to every dependent dataset:
axis_master_version
domain_master_version
key_master_version
network_master_version
module_registry_version

Any file built against an older Axis version must fail validation.
Recommended rule:

```
if dataset.axis_master_version != active_axis_master.version:
    reject_dataset(
        reason="AXIS_MASTER_VERSION_MISMATCH"
    )
```

Q2. Is there an official old-to-new Axis conversion table?
Developer question
The Product files use legacy Axis numbers. The team discovered that the current module-to-axis map had been compiled directly from those old numbers.
For example, the legacy Product files contained mappings such as:
Old 16 = Liver Detoxification
Old 17 = Digestive Function
Old 18 = Microbiome Ecology
Old 19 = Gut Barrier
But the new Master defines:
A15 = Detoxification Capacity
A16 = Digestive–Absorptive Function
A17 = Microbiome Ecology
A18 = Gut Barrier and Mucosal Integrity
The developers also identified cases where the same number retained its number but changed its meaning entirely, such as old 27 versus current A27.
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Official answer
Do not use a blind numerical conversion as the Production source of truth.
The updated Official Module Registry will directly contain the correct current Axis codes for every module.
The legacy Product-file Axis column must not be used in Production.
Why numerical conversion is unsafe
The migration problem is not a simple “subtract one” operation.
There are three possible situations:
Number shifted, meaning retained
Number retained, meaning changed
Legacy concept no longer maps cleanly to one current Axis
For example:
Legacy Axis 27 = Metabolic Balance
Current A27 = Glycation and Carbonyl Stress
A purely numerical migration would silently create a clinically incorrect mapping.
Correct migration process
Each legacy mapping must be reviewed semantically, using:
Product composition
Intended clinical role
Known biological mechanisms
Research evidence
Current 39-Axis definitions
Current 180-Network definitions
Then the updated Registry should directly store the new codes.
Source-of-truth rule

Production source of truth:
Updated Official Module Registry

Not a production source:
Legacy Product-file Axis Mapping
Role of a conversion table
A legacy-to-current conversion table may still be created for:
Migration support
Data lineage
Historical record interpretation
Audit
Detecting obsolete mappings
It must not automatically override the reviewed module mapping.
Recommended migration schema:
legacy_axis_number
legacy_axis_name
candidate_current_axis_codes[]
conversion_type
conversion_confidence
manual_review_required
reviewed_current_axis_codes[]
reviewed_by
review_date
notes
Possible conversion_type values:
DIRECT_EQUIVALENT
NUMBER_SHIFT
SEMANTIC_SPLIT
SEMANTIC_MERGE
MEANING_CHANGED
NO_DIRECT_EQUIVALENT
Registry validation rule
Each module should carry:
axis_master_version
mapping_status
mapping_source
mapped_by
clinically_reviewed_by
review_date
Allowed mapping statuses:
UNREVIEWED
PROVISIONAL
CLINICALLY_REVIEWED
APPROVED
DEPRECATED
Production should accept only:
APPROVED
or, during controlled testing:

CLINICALLY_REVIEWED
Q3. Does the 180 Living Biological Networks Master exist?
Developer question
The engine currently moves directly from Axes to Modules. However, TSPI philosophy states that the system should reason through biological networks rather than treating diseases as the primary reasoning units. The developers therefore asked whether a loadable 180-Network Master exists.
Official answer
The 180 Living Biological Networks are a core TSPI layer and must be treated as mandatory architecture, not an optional future enhancement.
The development team should build the complete Network loader and graph schema now.
Where the final Network Master is not yet fully structured, the system may use a clearly labelled placeholder dataset, but:
Placeholder Networks must not be treated as clinically authoritative.
The AI must not invent Network names.
Unresolved Network hypotheses must not enter deterministic scoring.
Correct reasoning architecture
The system should support bidirectional reasoning.
Clinical inference direction
Symptoms and history

→ Clinical phenotype

→ Candidate biological networks

→ Candidate Axes

→ Root-driver analysis

→ Harmonization options

Biological explanatory direction
Network dysregulation

→ Symptoms

→ Laboratory changes

→ Structural changes

→ Disease expression

The philosophical distinction matters:
A disease label is useful for communication and safety.
A disease label is not the core biological reasoning unit.
Living Networks are the dynamic reasoning units.
Axes are stable assessment dimensions that organize Networks.
The Gap Analysis correctly identifies the absence of the 180-Network layer as an architectural gap.
Recommended Network Master schema
network_code
network_number
network_name_en
network_name_th

definition
primary_axis_code
secondary_axis_codes[]
domain_code
primary_key_code
secondary_key_codes[]
network_type
upstream_network_codes[]
downstream_network_codes[]
feedback_relationships[]
accepted_symptoms[]
accepted_symptom_clusters[]
accepted_markers[]
accepted_clinical_features[]
accepted_exam_findings[]
accepted_imaging[]
accepted_omics[]
clinical_outputs[]
exclusion_rules[]
minimum_evidence
evidence_grade
framework_version
status
Recommended network_type values include:
SIGNALING
METABOLIC
IMMUNE
STRUCTURAL
TRANSPORT
CLEARANCE
REPAIR
REGULATORY
VASCULAR
NEUROENDOCRINE
MICROBIOME
BARRIER
REPRODUCTIVE
SENSORY
BIOMECHANICAL
INTEGRATIVE
Anti-hallucination rule
Only Networks present in the official Network Registry may be shown as established Networks.

```
if network_code not in official_network_registry:
    evidence_status = HYPOTHESIS
    scoring_effect = 0
    display_label = UNRESOLVED_NETWORK_HYPOTHESIS
```

An LLM may propose a candidate hypothesis in the narrative layer, but it must never create an official Network identity or score.

Q4. Is there a Symptom-to-Axis and Symptom-to-Network Dictionary?

Developer question

The current engine stores symptoms as free text and derives most Axis scoring from laboratory values. This conflicts with TSPI philosophy, in which symptoms are biological evidence and may be present before laboratory or structural changes become measurable.

Official answer

A formal Symptom-to-Axis and Symptom-to-Network Dictionary is required.

Symptoms must be treated as first-class biological evidence, not merely as narrative text.

However, the engine must never use:

one symptom = one Axis

or:

one symptom = one Network

The correct model is probabilistic and differential:

Symptom

→ multiple candidate Networks

→ multiple candidate Axes

→ weighting by context and co-symptoms

→ differential biological model

Required symptom structure

Each symptom should be recorded with structured attributes:

symptom_code
canonical_name_en
display_name_th
onset
duration
frequency
severity
progression
location
quality
triggering_factors
relieving_factors
meal_relationship
bowel_relationship
sleep_relationship
stress_relationship
activity_relationship
circadian_pattern
functional_impact
associated_symptoms[]
negative_findings[]
red_flags[]

Suggested base weighting

Clinical relationship

Initial weight
Weak / nonspecific association
0.25
Moderate association
0.50
Strong association
0.75
Pattern-defining association
1.00
These weights must be adjusted by:
Co-symptom clusters
Timing
Triggers
Relieving factors
Objective findings
Negative tests
Known confounders
Longitudinal consistency
Example: bloating
Bloating alone
Bloating is nonspecific.
The engine should create several low-confidence candidate Networks, potentially involving:
Digestion
Motility
Microbiome fermentation
Mucosal barrier
Autonomic regulation
Food intolerance
Fluid or gas handling
Bloating + early satiety + postprandial belching
Increase weight for:
A16 — Digestive–Absorptive Function
and relevant upper gastrointestinal motility and digestive Networks.
Bloating + constipation + abnormal stool pattern
Increase weight for:
A16 — Digestive–Absorptive Function
A17 — Microbiome Ecology
A19 — Colon Biofilm and Mucosal Burden
plus intestinal motility and enteric nervous-system Networks.
Bloating worsened by psychological stress
Increase the probability of:
A34 — Neuro–Sleep–Stress–Brain Clearance
plus autonomic digestive Networks and CELA-related regulation.
This is the type of contextual weighting the developers referenced in Q4.
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Recommended Dictionary schema
symptom_code
symptom_name_en

symptom_name_th
candidate_axis_code
candidate_network_code
base_weight
weight_class
required_co_symptoms[]
supporting_co_symptoms[]
contradicting_features[]
trigger_rules[]
temporal_rules[]
negative_test_rules[]
red_flag_links[]
minimum_evidence
version
status

Important negative-test principle

A normal test must not erase a symptom.

For example:

Normal gastroscopy

≠ normal digestion

≠ normal motility

≠ normal enteric signaling

≠ normal microbiome

A normal test narrows the differential. It does not prove that the entire biological Network is normal.

Q5. What is the official Red-Flag list?

Developer question

The developers correctly identified that the current system lacks a Red-Flag layer before Network and Module reasoning. This is a safety-critical gap.

Official answer

Red-Flag Screening is mandatory and must run before any TSPI Network, Axis or Module recommendation.

The required workflow is:

Symptoms and history

→ Red-flag screening

→ Standard medical differential and required investigation

→ Clinical phenotype construction

→ Network mapping

→ Axis integration

→ Module consideration

Three Red-Flag action classes

Class 1 — Emergency referral

The system should immediately block the normal TSPI recommendation flow and direct emergency evaluation.
Examples include:
Sudden unilateral weakness
New speech disturbance
Sudden severe ataxia
Seizure
Loss of consciousness
Severe chest pain
Severe respiratory distress
Shock
Major active bleeding
“Worst headache of life”
Anaphylaxis
Severe hypoglycemia
Critical electrolyte abnormality
Severe hypoxemia
Recommended action:
action = EMERGENCY_REFERRAL
release_block = true
module_plan_allowed = false
Class 2 — Urgent medical assessment
Examples include:
Persistent vomiting
Black stool
Bloody stool
New dysphagia
Severe abdominal pain
New jaundice
High fever with altered mental status
Severe anemia
Unexplained significant weight loss
New mass
New focal neurological symptoms
Symptomatic arrhythmia
Very low blood pressure with symptoms
Recommended action:
action = URGENT_MEDICAL_ASSESSMENT
release_block = true
module_plan_requires_clinician_override = true
Class 3 — Standard diagnostic work-up before primary TSPI recommendation
Examples include:
New symptoms in an older patient
Persistent unexplained abnormalities
Change in a long-standing symptom pattern
Strong family history of cancer
Organ dysfunction of unclear cause
Persistent abnormal biomarker

Symptoms unresponsive to reasonable initial care
Potential malignancy, bleeding or obstruction requiring exclusion
Recommended action:
action = STANDARD_WORKUP_FIRST
cause_specific_module_plan = blocked
supportive_plan = clinician_review_only
Recommended Red-Flag schema
red_flag_code
symptom_area
red_flag_name_en
red_flag_name_th
description
action_class
required_action
recommended_referral
release_block
module_plan_allowed
override_role
version
status
The initial production system should not rely on the LLM to determine Red Flags from unrestricted prose alone. The Red-Flag engine should be deterministic, based on structured inputs and clinically reviewed rules.
Q6. What is the official order of the 3 Keys, and which Domains map to each Key?
Official order
The official TSPI Key order is:
Key 1 — Metabolic Energy
Key 2 — Biological Dynamics
Key 3 — Biointegrity
Recommended stable codes:
KEY-A = Metabolic Energy
KEY-B = Biological Dynamics
KEY-C = Biointegrity
Keep the stable key_code separate from display_order.
For example:
```
{
  "key_code": "KEY-A",
  "official_name": "Metabolic Energy",
  "display_order": 1
}
```
This allows presentation order to change in the future without corrupting database relationships.
Meaning of the Keys
Key A — Metabolic Energy
Represents the capacity to generate, distribute and use biological energy.
Includes:
Nutrient utilization
Mitochondrial ATP generation

Metabolic flexibility
Redox support
Energetic reserve
Capacity to sustain adaptation
Key B — Biological Dynamics
Represents biological movement, signaling, exchange and adaptation.
Includes:
Immune dynamics
Circulation
Hormonal communication
Neural regulation
Microbiome interactions
Transport
Feedback loops
Adaptive change
Key C — Biointegrity
Represents the capacity to maintain coherent biological identity, structure and long-term stability.
Includes:
Genomic stability
Epigenetic coherence
Tissue integrity
Organ reserve
Repair
Regeneration
Protection against malignant transformation
System-level biological coherence
Domain-to-Key mapping model
A Domain is not biologically isolated to one Key. Therefore, the database should support:
one Primary Key
zero to two Secondary Keys
The Primary Key represents the dominant organizational principle, not the Domain's only biological relationship.
Recommended TSPI Domain-to-Key Master v1.0
Domain
Primary Key
Secondary Keys
D1 Immune–Infection Governance
Biological Dynamics
Biointegrity, Metabolic Energy
D2 Metabolic–Energy Core
Metabolic Energy
Biological Dynamics, Biointegrity
D3 Genomic Programming
Biointegrity
Metabolic Energy, Biological Dynamics
D4 Detoxification and Cellular Defense
Metabolic Energy

Biointegrity, Biological Dynamics
D5 Digestive–Gut Ecosystem
Biological Dynamics
Metabolic Energy, Biointegrity
D6 Vascular–Circulation
Biological Dynamics
Metabolic Energy, Biointegrity
D7 Repair–Regeneration
Biointegrity
Metabolic Energy, Biological Dynamics
D8 Musculoskeletal and Structural Integrity
Biointegrity
Metabolic Energy, Biological Dynamics
D9 Endocrine Network
Biological Dynamics
Metabolic Energy, Biointegrity
D10 Neurological Regulation
Biological Dynamics
Metabolic Energy, Biointegrity
D11 Organ Resilience
Biointegrity
Metabolic Energy, Biological Dynamics
D12 Oncology and System-Level Regulation
Biointegrity
Biological Dynamics, Metabolic Energy
Recommended schema:
domain_code
domain_name
primary_key_code
secondary_key_codes[]
relationship_weights[]
framework_version
status
Initial relationship weights may remain descriptive until clinically validated. The system should not invent numerical Key weights without an approved Master.
PART 2 — Design Clarifications
Q7. How should the four required dimensions be calculated?
The developers asked whether the following four concepts are different:
Evidence Level
Confidence Level
Data Completeness
Biological Uncertainty
They are different and must be stored separately.
1. Evidence Level
“Evidence Level” should itself be separated into two fields.
A. Evidence Status
Evidence Status tells us how the statement was produced.
Allowed values:

MEASURED
DERIVED
CLINICAL_INFERENCE
HYPOTHESIS
NOT_AVAILABLE
Definitions:
MEASURED: directly obtained from laboratory, imaging, examination, validated device or structured patient report.
DERIVED: calculated from measured data using a defined formula.
CLINICAL_INFERENCE: inferred from a clinically meaningful pattern.
HYPOTHESIS: biologically plausible but not sufficiently supported for scoring.
NOT_AVAILABLE: required evidence is absent.
B. Evidence Grade
Evidence Grade tells us how strong the scientific support is for the claimed relationship.
Allowed values:
A
B
C
D
Evidence Status and Evidence Grade are not interchangeable.
Example:
Measured serum marker:
Evidence Status = MEASURED

Marker-to-Network relationship supported mainly by animal studies:
Evidence Grade = C
Scoring rule
HYPOTHESIS must never raise:
Axis score
Network score
NSS
Module Match score
A Hypothesis may only:
Generate a follow-up question
Suggest an investigation
Remain in a differential model
Be displayed with explicit uncertainty
2. Confidence Level
Confidence represents how strongly the available evidence supports the conclusion for this patient.
Scale:
0.00–1.00
Suggested interpretation:
Score
Label
0.90–1.00
Very High
0.75–0.89

High
0.50–0.74
Moderate
0.25–0.49
Low
0.00–0.24
Very Low
Confidence should reflect:
Evidence consistency
Evidence quality
Pattern specificity
Number of independent evidence sources
Longitudinal consistency
Measurement reliability
Contradicting evidence
Relevant confounders
Confidence must not be based only on the number of data points.
3. Data Completeness
Data Completeness measures how much of the expected information for a conclusion is available.
It should not simply be:
markers present ÷ markers expected
because all evidence items are not equally important.
Use a weighted formula:
Data Completeness =
Σ weights of available expected evidence
÷
Σ weights of all expected evidence
Evidence items should be classified as:
CORE
SUPPORTING
OPTIONAL
Example weights:
CORE = 3
SUPPORTING = 2
OPTIONAL = 1
If all optional data are present but a critical Core Marker is missing, Data Completeness should not appear high.
4. Biological Uncertainty
Biological Uncertainty represents how many biologically plausible explanations remain after considering the available evidence.
It differs from Confidence.
Confidence:
How strongly do we support the current conclusion?

Biological Uncertainty:
How many important alternative biological models remain plausible?
Example:

A patient may have complete and reliable symptom data, but three different Network combinations may explain the pattern equally well.
In that case:
Data Completeness = high
Evidence quality = good
Confidence in one single model = moderate
Biological Uncertainty = high
Recommended scale:
0.00–1.00
where:
0 = minimal uncertainty
1 = very high uncertainty
Suggested display labels:
Score
Label
0.00–0.24
Low
0.25–0.49
Moderate
0.50–0.74
High
0.75–1.00
Very High
Biological Uncertainty increases when:
Several candidate Networks fit similarly
Key discriminating tests are absent
Data conflict
Strong confounders are present
Temporal behavior is unclear
A symptom is highly nonspecific
A causal direction cannot yet be distinguished
Q8. Are sub-axes used for scoring?
Official answer
Yes, but sub-axes and Parent Axes have different functions.
Sub-axes are used for:
Evidence organization
Detailed scoring within an Axis
Mechanistic explanation
Follow-up test selection
Longitudinal tracking
Modules are primarily mapped to Parent Axes A1–A39.
Correct structure:
Evidence

→ Sub-axis

→ Parent Axis score

→ Network reasoning

→ Module matching

The team’s interpretation is therefore correct:
Sub-axes are not required as the main Module-mapping level, but they may be used for clinical evidence and scoring within a Parent Axis.
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Example
A10 may have sub-axes:
A10A — Protein Folding and Chaperones
A10B — ER Stress and UPR
A10C — Autophagy–Lysosome Clearance
A10D — Proteotoxic Burden
A Module may map to:
Parent Axis = A10
and additionally carry optional mechanism tags:
sub_axis_tags = [A10A, A10C]
These tags can refine ranking, but should not create duplicate Product records.
Q9. Can the latest Master Dictionary be loaded directly as the Axis Master?
Official answer
The latest Master Dictionary should be used as the canonical skeleton, but it should be transformed into a structured Axis Master before Production use.
The Axis Master should include:
axis_code
axis_number
axis_name_en
axis_name_th
domain_code
definition
sub_axes[]
accepted_markers[]
accepted_symptoms[]
accepted_symptom_clusters[]
accepted_clinical_features[]
accepted_exam_findings[]
accepted_imaging[]
accepted_omics[]
accepted_networks[]
exclusion_rules[]
must_not_infer_from[]
minimum_evidence
scoring_method
critical_override_rules[]
framework_version
status
Minimum Evidence
Minimum Evidence should not be identical for every Axis, because some Axes can be reasonably estimated from symptoms and routine tests, whereas others require more specific evidence.

Use a shared status framework, with Axis-specific criteria.
POSSIBLE
At least one compatible evidence source is present, but the finding is nonspecific or incomplete.
axis_status = POSSIBLE
confidence = low
score_effect = limited
PROBABLE
At least two reasonably independent and concordant evidence sources are present.
Examples:
Symptom cluster + laboratory marker
Examination finding + imaging
Multiple concordant laboratory markers
Longitudinal symptom pattern + physiological measurement
HIGH_CONFIDENCE
Multiple evidence layers are concordant, or a highly specific finding is present with appropriate clinical correlation.
NOT_ASSESSED
Insufficient usable evidence exists.
axis_status = NOT_ASSESSED
reason = INSUFFICIENT_DATA
No data must never be interpreted as normal.
This directly addresses the Gap Analysis finding that unscored Axes are currently absent rather than explicitly represented as NOT_ASSESSED.
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Q10. Confirm the Module Match formula
Official Version 1 formula
The proposed formula is approved as a starting model:
30% Axis Match
25% Network Match
15% Evidence Grade
10% Patient Phenotype Fit
10% Safety
5% Treatment-Step Compatibility
5% Historical Response
Name it:
TSPI Module Match v1.0
This formula should remain configurable and version-controlled rather than hard-coded permanently.
Safety is both a Gate and a Score
The 10% Safety component applies only after the Module passes mandatory exclusion checks.
Safety Gate
If an absolute contraindication or major unacceptable interaction exists:
module_status = EXCLUDED
module_match_score = not_calculated
Do not merely subtract 10 points.
Safety ranking

For Modules that pass the Gate, the Safety component may rank:
Lower-risk Module
Better interaction profile
Better fit with organ function
Better fit with age or comorbidities
What does “excessive Module burden” mean?
Excessive Module burden refers to a plan containing so many Modules that it:
Duplicates mechanisms
Increases interaction risk
Increases pill burden
Reduces adherence
Prevents clear outcome attribution
Violates treatment-step priorities
Makes the learning loop unreliable
Creates an unrealistic clinical plan
Because Modules often map broadly across several Axes, a severe patient may match dozens of Modules. Matching does not mean prescribing.
Recommended selection limits
Typical case
Core Modules: 1–3
Supporting Modules: 0–3
Typical total maximum: 6
Complex case
7–8 Modules may be considered
Requires senior clinical approval
More than 8
Requires:
Explicit clinical justification
Duplicate-mechanism review
Interaction review
Senior override
Documentation
This is a recommendation ceiling, not an absolute prescribing law. The physician retains authority.
Mechanism de-duplication
The engine should identify Modules with:
Same primary mechanism
Same dominant Axis pattern
Same bowel-regulating role
Same safety burden
Redundant treatment-step function
The engine should select the highest-ranked representative unless the physician overrides.
Evidence Grade A–D
Grade A
Strong, directly relevant human clinical evidence, preferably reproduced or supported by high-quality trials.
Grade B

Human observational, pilot or early clinical evidence, supported by coherent preclinical evidence.
Grade C
Animal, cellular, proteomic, mechanistic or network-pharmacology evidence with biological plausibility.
Grade D
Traditional use, expert consensus or early hypothesis with limited direct experimental support.
Evidence Grade should preferably be stored at the level of:
Module
+ biological claim
+ target Axis or Network
not only as one global grade for the entire Product.
A Module may therefore have:
Grade B evidence for inflammation
Grade C evidence for mitochondrial function
Grade D evidence for a proposed neurological mechanism
Required reason_not_selected
Every matched but unselected Module should have a structured reason, such as:
CONTRAINDICATED
MAJOR_INTERACTION
DUPLICATE_MECHANISM
INSUFFICIENT_EVIDENCE
INSUFFICIENT_PATIENT_DATA
NOT_STEP_COMPATIBLE
EXCESSIVE_MODULE_BURDEN
LOWER_RANKED_THAN_SELECTED_ALTERNATIVE
PHYSICIAN_EXCLUDED
DIAGNOSTIC_WORKUP_REQUIRED_FIRST
The Gap Analysis correctly identifies the lack of a structured Module Match formula and reason_not_selected as a current gap.
TSPI_Phase3_Expert_Feedback_Gap_Analysis_260714_075629 (1).pdf
Q11. What should the Adaptive Biological Intelligence Loop learn?
Official answer
Network learning does not replace Axis learning. It sits above it.
The AI should learn at three levels.
Level 1 — Patient-specific Axis adaptation
Tracks how individual Axis scores change over time in one patient.
Examples:
A1 inflammatory load decreases
A8 mitochondrial function improves
A17 microbiome evidence becomes stronger
A35 organ reserve declines
This learning is patient-specific and bounded.
Level 2 — Network behavior learning
Tracks how Networks influence one another across time.
Examples:
Enteric Motility Network improves

→ bowel frequency improves within 1–2 weeks

Microbiome Resilience Network improves

→ inflammatory burden decreases over 4–8 weeks

Mitochondrial Bioenergetic Network improves

→ fatigue improves before exercise tolerance

Autonomic Regulation improves

→ digestive symptoms and sleep improve together

This is the meaning of learning “the behavior of biological Networks.”
The AI should record:
Which Network changed first
Which downstream Network changed later
Time to response
Magnitude of response
Whether the relationship repeated
Whether confounders were present
Level 3 — Whole Patient Biological Model learning
The engine should maintain a versioned, living patient model.
Patient Biological Model v1

→ Patient Biological Model v2

→ Patient Biological Model v3

Every new clinically meaningful input should reconstruct the current model.
New evidence may:
CONFIRM
REFINE
REDIRECT
OVERRIDE
CONTRADICT
a previous hypothesis.
The old model must not be deleted. It must remain available in the audit history.
Minimum learning record
patient_model_version
baseline_network_state
baseline_axis_state
intervention_modules[]
dose
duration
adherence
lifestyle_interventions[]
concurrent_medications[]
concurrent_treatments[]
followup_network_state
followup_axis_state

symptom_response
laboratory_response
imaging_response
adverse_response
time_to_response
confounders[]
physician_interpretation
learning_approval_status
Population-level learning rule
The AI must not automatically modify the Global Clinical Model.
It may only generate:
PROPOSE_MODEL_UPDATE
A proposed update must pass:
Minimum sample size
Data-quality review
Confounder review
Statistical analysis
Clinical review
External or retrospective validation
Version approval
Only then may a new Model Version enter Production.
This corrects the current noncompliant global auto-recalibration behavior identified in the Gap Analysis.
TSPI_Phase3_Expert_Feedback_Gap_Analysis_260714_075629 (1).pdf
Q12. What should a Network Harmonization Plan contain?
A Network Harmonization Plan should not be a simple Product list.
It should communicate:
What is known
What remains uncertain
Which Networks are likely upstream
Which Networks are compensating
Which Axes are active
Why particular Modules are considered
How the patient will be monitored
Recommended structure:
Section 1 — Patient Biological Context
Include:
Main symptoms
Timeline
Triggering factors
Relieving factors
Medical history
Current medications
Current Modules
Lifestyle context
Available evidence
Missing evidence
Section 2 — Red-Flag and Standard Medical Assessment

Include:
Red Flags present or absent
Standard medical differentials
Conditions requiring exclusion
Required referrals
Required investigations
Limits of current TSPI interpretation
Section 3 — Clinical Phenotype
Include:
Symptom clusters
Functional patterns
Temporal patterns
Trigger-response relationships
Adaptive capacity
Compensatory patterns
Functional impact
Section 4 — Differential Network Model
List multiple candidate Networks where appropriate.
For each Network show:
Evidence Status
Confidence Level
Data Completeness
Biological Uncertainty
Supporting evidence
Contradicting evidence
Missing discriminating evidence
Do not force one Root Network when the evidence supports several possibilities.
Section 5 — Root Driver Network Analysis
Classify Networks as:
CONFIRMED_ROOT_DRIVER
PROBABLE_ROOT_DRIVER
POSSIBLE_ROOT_DRIVER
NOT_YET_ASSESSABLE
Use “Root Driver” as a current biological model, not an absolute claim.
Section 6 — Secondary and Compensatory Networks
Identify Networks that may be:
Downstream consequences
Adaptive compensations
Protective responses
Overloaded compensations
Potential future failure points
A compensatory Network should not automatically be suppressed. The system must first determine whether it is harmful, protective or both.
Section 7 — Axis Integration
Show the relationship between Networks and the official 39 Axes.
Axis statuses may include:
PRIMARY_ACTIVE_AXIS
SECONDARY_ACTIVE_AXIS

MONITORING_AXIS
COMPENSATORY_AXIS
NOT_ASSESSED
Section 8 — Harmonization Priorities
Priority 1
Likely Root-driver Networks
Priority 2
Supporting, downstream or compensatory Networks requiring support
Priority 3
Resilience, maintenance and relapse-prevention Networks
This prevents broad Axis matches from generating an unstructured list of dozens of Products.
Section 9 — Module Options
Only official Modules from the Registry may appear.
Each Module should show:
H-Code
Canonical English name
PhytoCore Code, if available
Axis Match score
Network Match score
Evidence Grade
Phenotype fit
Safety status
Treatment-step compatibility
Historical response
Dose rule
Reason selected
Reason not selected for alternatives
No invented Module names are permitted.
Section 10 — Lifestyle and Environmental Harmonization
May include:
Nutrition
Sleep
Circadian regulation
Physical movement
Breathing practices
CELA regulation
Microbiome support
Autonomic support
Hydration
Environmental exposure
Social and emotional factors
These are not secondary “tips.” They are potential Network-level interventions.
Section 11 — Expected Biological Response
State expected directions without guaranteeing an outcome.
For example:
Expected early response:
Improved bowel regularity and reduced post-meal discomfort

Expected intermediate response:
Improved digestive tolerance and lower inflammatory symptom burden

Expected later response:
Improved network resilience and reduced relapse frequency
Section 12 — Monitoring Plan
Specify:
Symptoms to track
Markers to repeat
Axes to reassess
Networks to reassess
Timing
Escalation conditions
Criteria to continue, adjust or stop the plan
Section 13 — Biological Model Update
State when the next Patient Biological Model will be generated.
For example:
Rebuild patient model:
2 weeks after treatment initiation,
or earlier if a critical new symptom or laboratory result appears.
Q13. Confirm the four patient-communication levels
Official answer
The following structure is confirmed:
Confirmed
Probable
Not Yet Known
Suggested Next Tests
This structure should be used in both physician-facing and patient-facing reports, with different technical detail.
Example
Confirmed
The CBC demonstrates a microcytic, hypochromic pattern.
Probable
Iron deficiency and thalassemia trait are important possibilities.
Not Yet Known
The cause has not yet been confirmed.
Suggested Next Tests
Ferritin, iron profile and hemoglobin typing are recommended.
Can a Module be recommended before the cause is confirmed?
Do not use the absolute rule:
No Module may ever be displayed before a cause is confirmed.
That rule is too rigid for Fundamental Medicine because low-risk Network support may sometimes be appropriate while the differential remains open.
Use the more precise rule below.
Cause-specific Module recommendation is blocked when:
The cause remains unclear
A dangerous diagnosis has not been excluded

The Module may interfere with diagnosis
The Module may mask an important symptom
The risk profile is unacceptable
A standard medical work-up is required first
Supportive Network Harmonization may be shown when:
Red-Flag Screening is passed
It does not replace standard investigation
It is low risk
It does not interfere with diagnosis
Uncertainty is explicitly displayed
Follow-up testing is included
A physician approves the plan
Recommended display wording:
This is a supportive Network Harmonization option.
It is not a confirmed cause-specific treatment and does not replace the recommended diagnostic work-up.
PART 3 — Master Datasets Still Required
The development team may build schemas and loaders immediately for:
Official Module Registry
Marker Registry
Laboratory Cut-off Registry
Axis Master
Domain-to-Key Master
180-Network Master
Symptom-to-Axis Dictionary
Symptom-to-Network Dictionary
Red-Flag Registry
Evidence Grade Registry
NSS Coefficient Registry
Negative-Test Interpretation Registry
Each Master file should contain:
dataset_name
dataset_version
framework_version
effective_date
approval_status
reviewed_by
approved_by
change_log
supersedes_version
status
Recommended status values:
DRAFT
PROVISIONAL
CLINICALLY_REVIEWED
APPROVED
DEPRECATED
ARCHIVED

No Draft or Provisional dataset should silently enter clinical Production.

Final Developer Lock

The following decisions are now locked:

The latest “39 Axes, 12 Domains, 3 Keys” file is the canonical framework.

A10 is Proteostasis–Autophagy–Lysosome.

A37 is Oncologic Cell Fate.

A38 is Tumor Microenvironment and Metastatic Dynamics.

A39 is Genomic Regulation Homeostasis.

CELA is a Living Biological Network, not A39.

Key 1 is Metabolic Energy.

Key 2 is Biological Dynamics.

Key 3 is Biointegrity.

The 180 Networks are a mandatory reasoning layer.

Legacy Product-file Axis mappings are not a Production source of truth.

The updated Module Registry must contain current Axis codes directly.

Sub-axes support evidence and scoring; Modules primarily map to Parent Axes.

Symptoms are first-class biological evidence.

Red-Flag Screening must occur before Network or Module reasoning.

Safety contraindications operate as hard exclusion gates.

Global model learning is propose-only and requires clinical approval.

Every patient-facing recommendation requires physician approval.

Every conclusion must expose Evidence Status, Confidence, Data Completeness and Biological Uncertainty.

No missing data may be interpreted as normal.

No LLM may independently create an Axis score, Network identity or Module name.

Every report and dataset must be version-stamped.

The existing deterministic scoring pipeline, official Module resolution, physician approval gate and versioned architecture should be retained. The engine does not need to be rebuilt from zero. It needs the latest ontology frozen, the legacy mappings removed, and the Clinical Phenotype, Red-Flag and 180-Network layers added around the existing foundation. This aligns directly with the Gap Analysis conclusion that the major task is to freeze the Master ontology and make every clinical calculation flow through authoritative data rather than invented or legacy mappings.